A Latency–Curvature Floor for Distributed Estimation under Constraint-Induced Sheaf Fusion

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Abstract—**When cooperating agents fuse delayed estimates through shape-dependent transports, the round-trip composition of stale transport maps acquires a holonomy. We prove that, to leading order in the staleness τ, the error-transport holonomy amplitude obeys $\|\operatorname{Log}\mathrm{Hol}\| = \tau^2\,\|[C_i, C_j]\| + O(\tau^3)$, with C_j the shape-conjugated error generators, vanishing identically under each of the theorem's hypotheses' negations ($\eta = 0$, $\xi = 0$, coincident shapes). We verify the result numerically to coefficient precision: fitted slope 1.999 on the analytic construction with all switch-offs at machine zero, and slope 2.000 with the m=2 coefficient ratio 1.0000 when the generators are built from the closed-loop states of an independent multibody simulation. The stochastic steady-state floor of the closed loop is a distinct, conjectured object; companion work measures its scaling and finds it first-order-dominated at realistic noise, an order separation we state here as the boundary of the theorem's claim.**

I. Introduction

[Author section: motivation from the companion theory — cable-coupled multi-agent estimation, the constraint channel, delayed fusion through shape-conjugated transports, and why the round trip is the natural object. Source: estimation_sheaf.tex §1.**]**

II. Setting and main result

[Author section: Theorem 7.2 stated verbatim with its tag — "(leading order, deterministic; sharp constants and the stochastic steady-state floor: [Conjectural])" — the hedge is load-bearing; plus the supporting BCH lemma (Lem. 7.1). Source: estimation_sheaf.tex §7.**]**

III. Construction and proof architecture

A. Conventions and the constraint trivialization

On $SE(2)$, $\operatorname{Exp}(\xi) = (R(\omega), V(\omega)v)$ for $\xi = (v, \omega)$ with the standard closed-form V, and $\mathrm{Ad}_{(R,t)} = \left[\begin{smallmatrix} R & Jt \\ 0 & 1 \end{smallmatrix}\right]$, $Jt = (t_y, -t_x)^\top$. An agent's channel shape is $s_j = (\sigma, \sigma_i)$ — its cable's direction in the load frame and in its own frame — and the taut cable of length l trivializes the agent pose:

$$m(\sigma, \sigma_i) = \big(R(\sigma - \sigma_i),\, l(\cos\sigma, \sin\sigma)\big), \qquad g_{\text{agent}} = g_L\, m(s), \tag{1}$$

with load-independent edge maps $g_i^{-1} g_j = m_i^{-1} m_j$. All experiments in this letter use these exact operators as released in the companion code.

B. Why delayed fusion transports errors by $e^{\tau C_j}$

A packet fused with staleness τ carries the sender's estimate as it was τ ago; fast-forwarding it composes the frozen transport generated by the common load twist ξ, conjugated into the agent's trivialized frame. The frozen error-transport generator of agent j is therefore

$$C_j(s_j; \xi) \;=\; \mathrm{Ad}_{m(s_j)}\, \mathrm{ad}_\xi\, \mathrm{Ad}^{-1}_{m(s_j)}, \tag{2}$$

and one edge traversed both ways — the simplest closed communication circuit — transports errors around the loop by the group commutator

$$\mathrm{Hol} \;=\; e^{\tau C_i}\, e^{\tau C_j}\, e^{-\tau C_i}\, e^{-\tau C_j}. \tag{3}$$

C. The leading term and its null directions

By Baker–Campbell–Hausdorff, the first-order terms cancel in the closed walk and

$$\operatorname{Log}\mathrm{Hol} \;=\; \tau^2\,[C_i, C_j] \;+\; O(\tau^3), \tag{4}$$

which is the object the main theorem bounds (Lem. 7.1 supplies the expansion; the theorem's statement and proof appear in Sec. II). Each hypothesis of the theorem corresponds to an exact null of the leading term, and each is a measurable switch-off: $s_i \equiv s_j \Rightarrow C_i = C_j \Rightarrow [C_i, C_j] = 0$ (the symmetric class, Cor. 7.3); $\xi = 0 \Rightarrow \mathrm{ad}_\xi = 0$; and the leading commutator is proportional to the load turn rate η, vanishing identically at $\eta = 0$ (verified to machine zero). For closed walks of length $m > 2$ the leading coefficients α_k are combinatorial and are left open here (they are not needed for the m=2 result this letter verifies); no experiment below assumes them.

D. Measurement protocol

The amplitude is measured as $\|\operatorname{Log}\mathrm{Hol}\|$ with dense matrix exponentials/logarithms (no leading-order shortcut), so the fitted slope and the coefficient ratio $\|\operatorname{Log}\mathrm{Hol}\|/\tau^2\|[C_i, C_j]\|$ are independent checks: the former tests the order, the latter the constant. The τ-dependence of Hol is analytic once the generators are fixed; the measured content of the verification is therefore the coefficient, the switch-offs, and the $O(\tau^3)$ departure at the top of the grid — stated so the slope itself is never presented as an independent discovery.

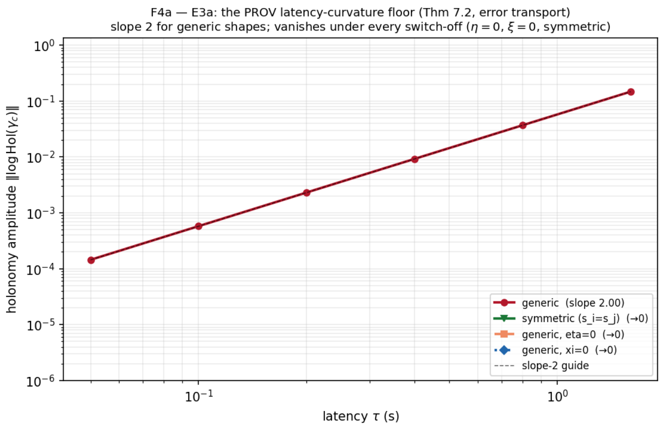

Fig. 1. Measured $\|\mathrm{Log\,Hol}\|$ vs. τ (analytic shapes). Generic pair: fitted slope 1.999 over $\tau \in [0.05, 1.6]\,$s; the coefficient ratio $\|\mathrm{Log\,Hol}\|/\tau^2\|[C_i, C_j]\| = 1.0000$ at the smallest τ. Switch-off arms ($s_i{=}s_j$; $\eta{=}0$; $\xi{=}0$) sit at machine zero ($\xi{=}0$ literally 0.0) — the effect dies exactly when the theorem requires it to.

TABLE I
SWITCH-OFF LADDER (TIER-1 ROWS; EXECUTED VALUES).

arm	prediction	measured
generic slope	2	1.999
coefficient ratio (m=2)	1	1.0000
$s_i \equiv s_j$	0	$< 10^{-15}$
$\eta = 0$	0	$< 10^{-17}$
$\xi = 0$	0	0.0

IV. The Measured Object

The m=2 round trip composes one edge's error transport both ways: $\mathrm{Hol} = e^{\tau C_i} e^{\tau C_j} e^{-\tau C_i} e^{-\tau C_j}$, with $C_j = \mathrm{Ad}_{m(s_j)}\,\mathrm{ad}_\xi\,\mathrm{Ad}^{-1}_{m(s_j)}$. Lemma 7.1 gives $\mathrm{Log\,Hol} = \tau^2[C_i, C_j] + O(\tau^3)$.

a) Independent-plant check.: Building the same generators from the closed-loop shape and twist *estimates* of a physics-complete multibody simulation (Drake; SAP contact; 12 m distance-constraint cables; five-vessel pentagon tow) gives slope 2.000, coefficient ratio 1.0000, the η=0 switch-off at machine zero, and a $31\times$ coefficient suppression for the achieved-ε symmetric class (exact cancellation remains the analytic result) — the theorem's object survives on states it was never derived from.

V. What the Theorem Does Not Claim

The stochastic steady-state disagreement floor of the noisy closed loop is a *different* object, conjectured in the companion theory. Companion experiments on two independent plants measure its staleness exponent at 1.08–1.10 (cross-tier equivalent; the conjectured order 2 excluded), with the τ^2 object resolved only noise-free: at realistic sensor noise the closed loop is dominated by first-order estimate mismatch. We state this boundary here so the theorem is cited for what it proves.

VI. Reproducibility

Every number above traces to a seeded, versioned driver; `campaign_replay.py` in the companion repository re-executes per-run probes that must match committed records exactly.